Foundations Plates

Stage-0 — The Seven Linear-Algebra Components of Hs

HUF-STD-003 · 2026-05-14

Per-country Foundations Plates for the 9-country EMBER corpus.
Each country: 2 pages — six-panel foundations grid + numeric verification table.

Foundations visualized (HUF-STD-003 §1-§7):

- §1 Symmetric matrix — variation matrix heatmap ($D \times D$)
- §2 Property of transpose — Helmert basis H + orthonormality check $HH^T = I$
- §3 Matrix decomposition — raw → CLR → ILR chain
- §4 Eigenvectors/eigenvalues — eigenvalue scree + cumulative variance
- §5 Spectral theorem — $\Sigma = Q \Lambda Q^T$ residual at IEEE floor
- §6 Spectral decomposition — orthonormal eigenbasis Q heatmap + rank-k explained
- §7 Visualization — the plate itself is this component

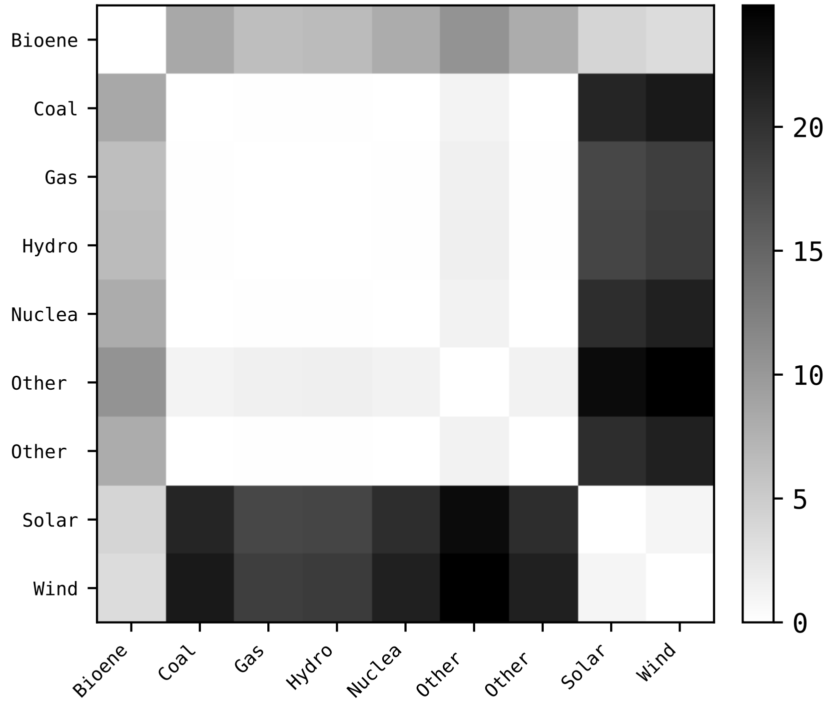
Stage 0 reads ONCE per dataset — the foundations characterize the data's geometry,
not its per-timestep state. Stage 1 plates (Section + Triplet) read per timestep.
Together they form the complete per-dataset reading.

P. Higgins, Rogue Wave Audio · AI Use Declaration in HUF-STD-003 · The foundations carry the bedrock.

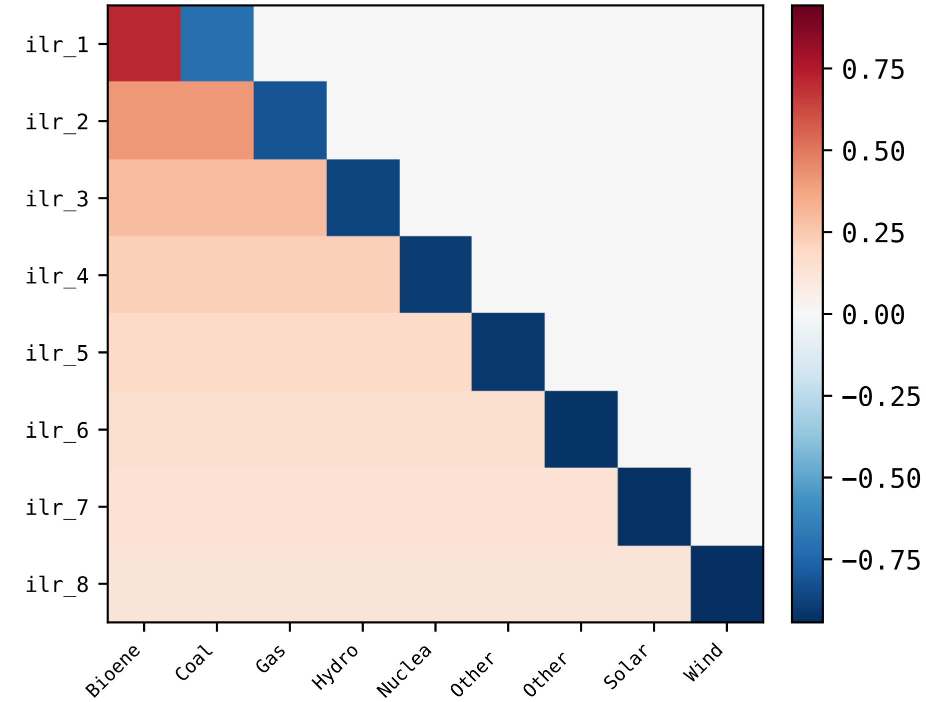
Foundations Plate (Stage 0) — EMB AUS Australia generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00



§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

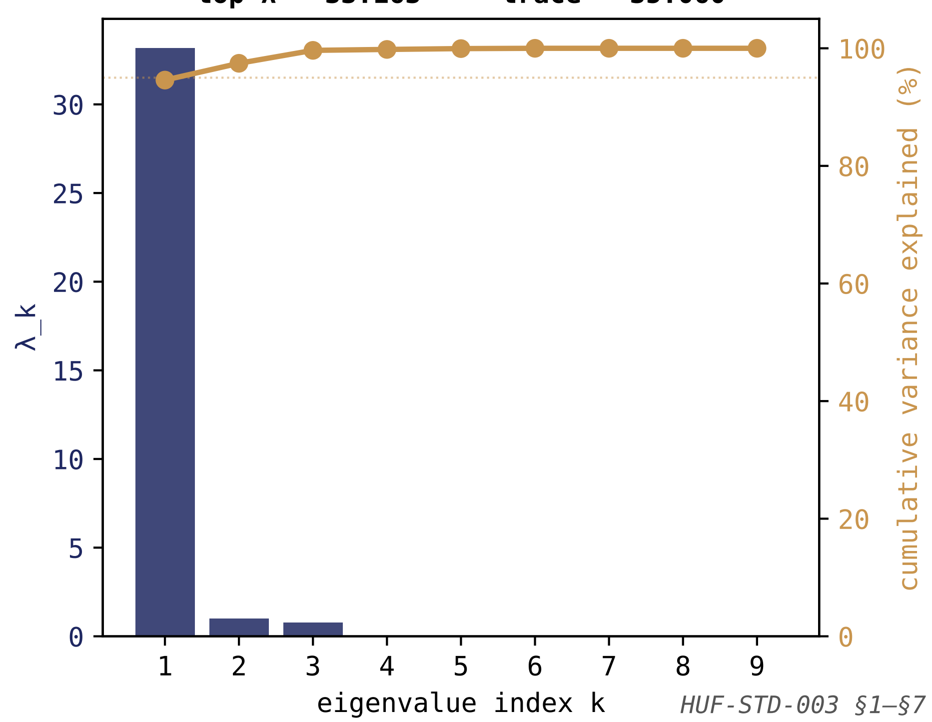


§3 Matrix Decomposition Chain

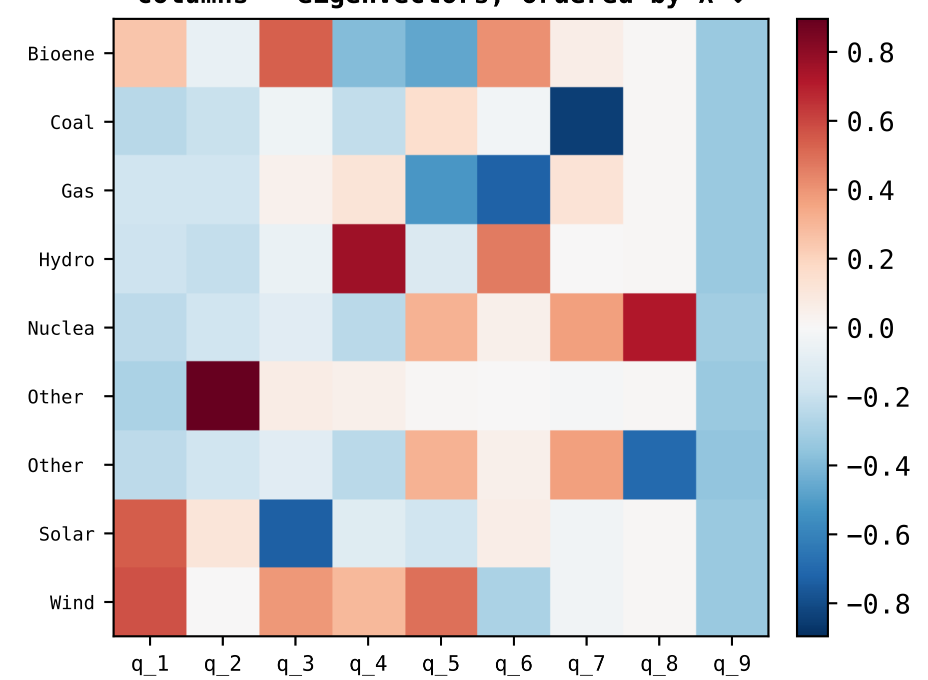
```
raw composition
     │  closure C(x) = x / Σx
     ▼
closed composition x ∈ Δ^{D−1}
     │  CLR: x → log x − ⟨log x⟩
     ▼
CLR coordinates (R^9)
     │  ILR: clr → clr @ H^T
     ▼
ILR coordinates (R^8)
```

- D = 9 carriers
- N = 26 timesteps
- each arrow is exact (modulo IEEE float)

§4 Eigenvalue scree (CLR covariance)
top λ = 33.163 · trace = 35.060



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓



§5 Spectral Theorem verification

Σ = Q Λ Q^T (theorem)
max|Σ − Q Λ Q^T| = 8.88e-15
(IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00 (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 1.33e-15

Rank-k variance explained:
k=1 → 94.6%
k=2 → 97.4%
k=3 → 99.6%
k=9 → 100.0%

Foundations — Numeric Verification — EMB AUS Australia generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

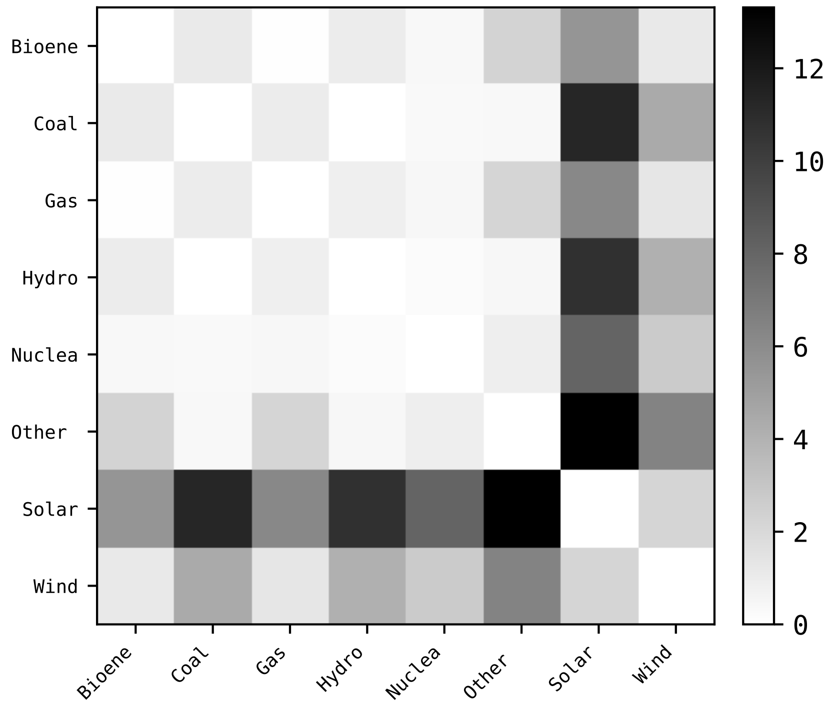
FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	3.333e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	33.1632	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	35.0596	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	8.882e-15	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	1.332e-15	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	94.59%	—
§6 Decomp	cumulative variance at k=2	97.45%	—
§6 Decomp	cumulative variance at k=3	99.65%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.

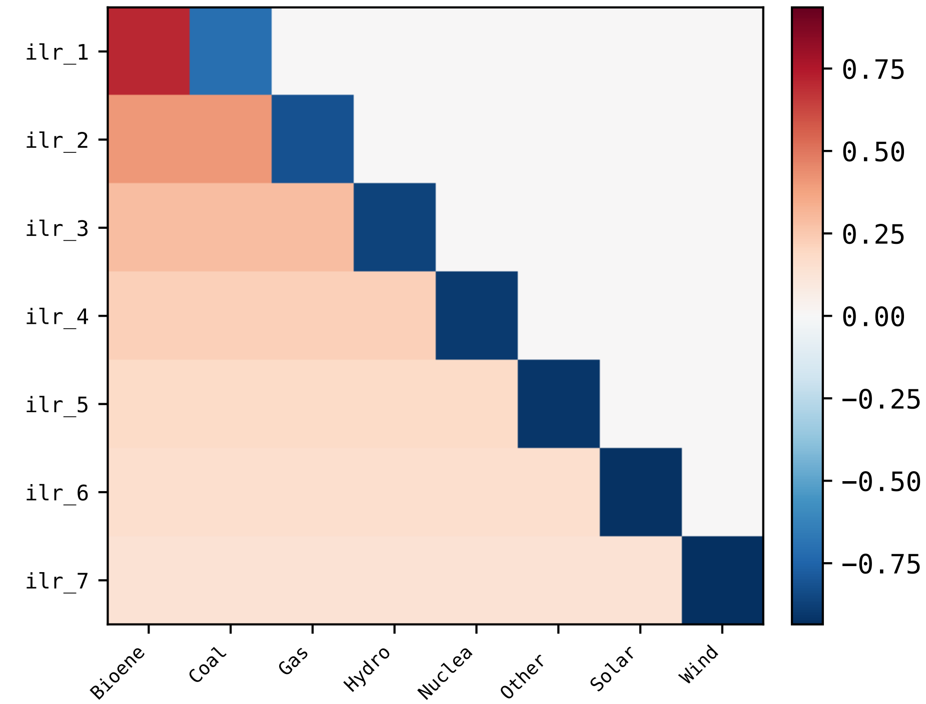
Foundations Plate (Stage 0) — EMB CHN China generation TWh.csv · D = 8 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00



§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

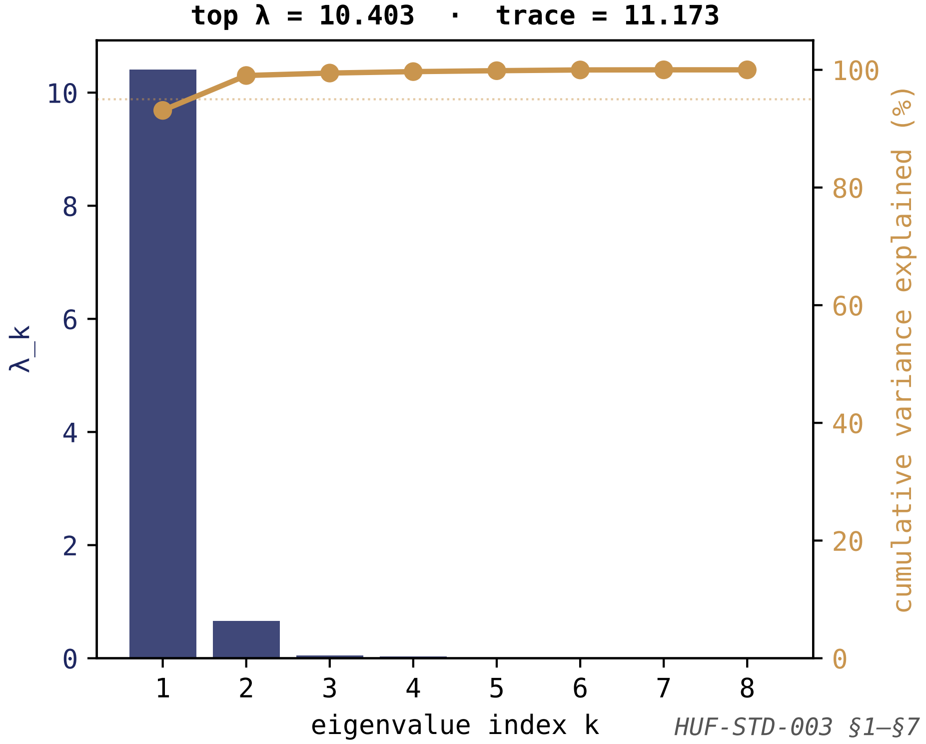


§3 Matrix Decomposition Chain

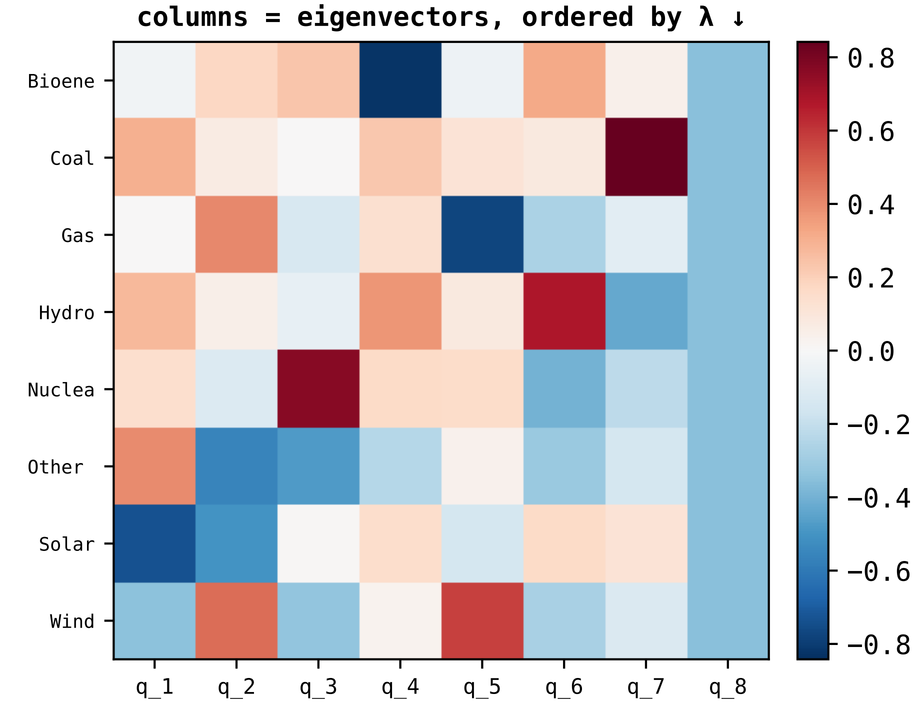
```
raw composition
     │  closure C(x) = x / Σx
     ▼
closed composition x ∈ Δ^{D−1}
     │  CLR: x → log x − ⟨log x⟩
     ▼
CLR coordinates (R^8)
     │  ILR: clr → clr @ H^T
     ▼
ILR coordinates (R^7)

 ·  D = 8 carriers
 ·  N = 26 timesteps
 ·  each arrow is exact
    (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 10.403 · trace = 11.173



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓



§5 Spectral Theorem verification

```
Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 4.00e-15
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 7.77e-16

Rank-k variance explained:
  k=1  →  93.1%
  k=2  →  99.0%
  k=3  →  99.5%
  k=8  →  100.0%
```

Foundations — Numeric Verification — EMB CHN China generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	1.250e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	8	= D (8)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	10.4029	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	11.1729	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	3.997e-15	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	7.772e-16	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	93.11%	—
§6 Decomp	cumulative variance at k=2	99.04%	—
§6 Decomp	cumulative variance at k=3	99.45%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
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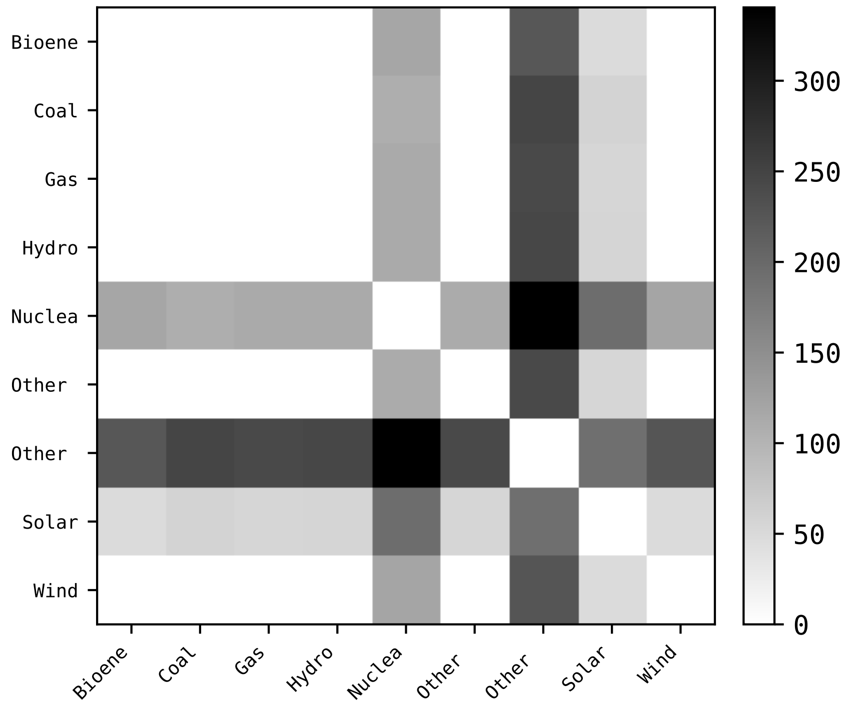
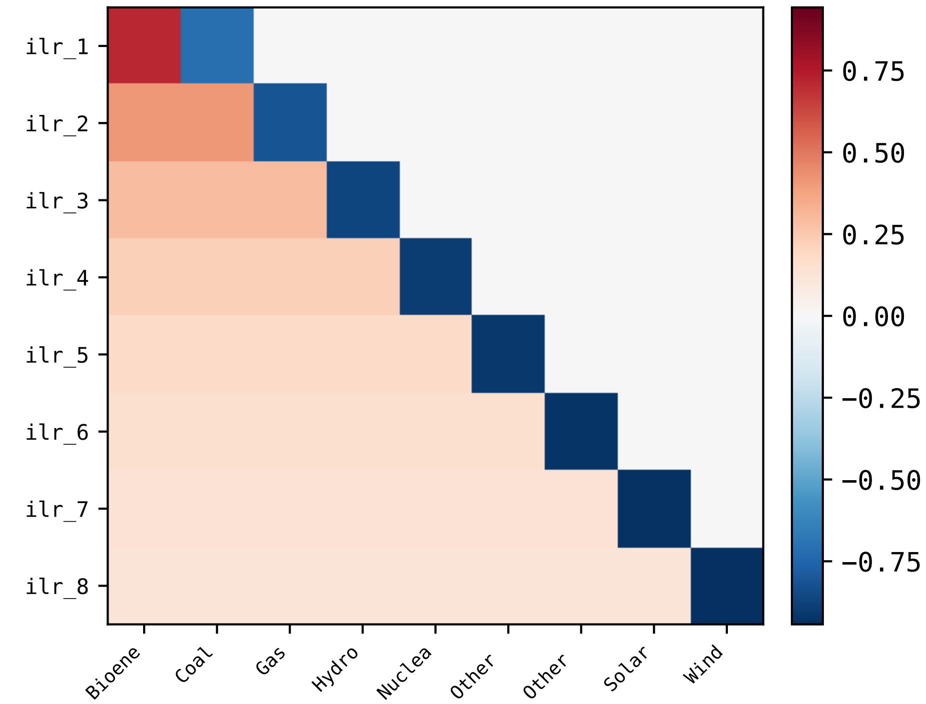
Foundations Plate (Stage 0) — EMB DEU Germany generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00



§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

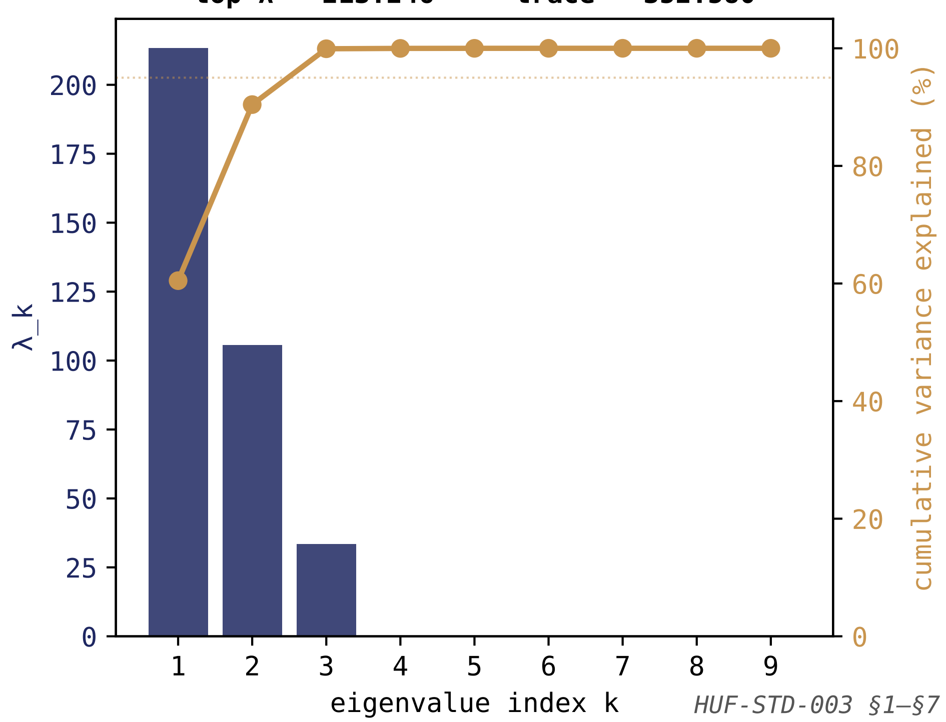


§3 Matrix Decomposition Chain

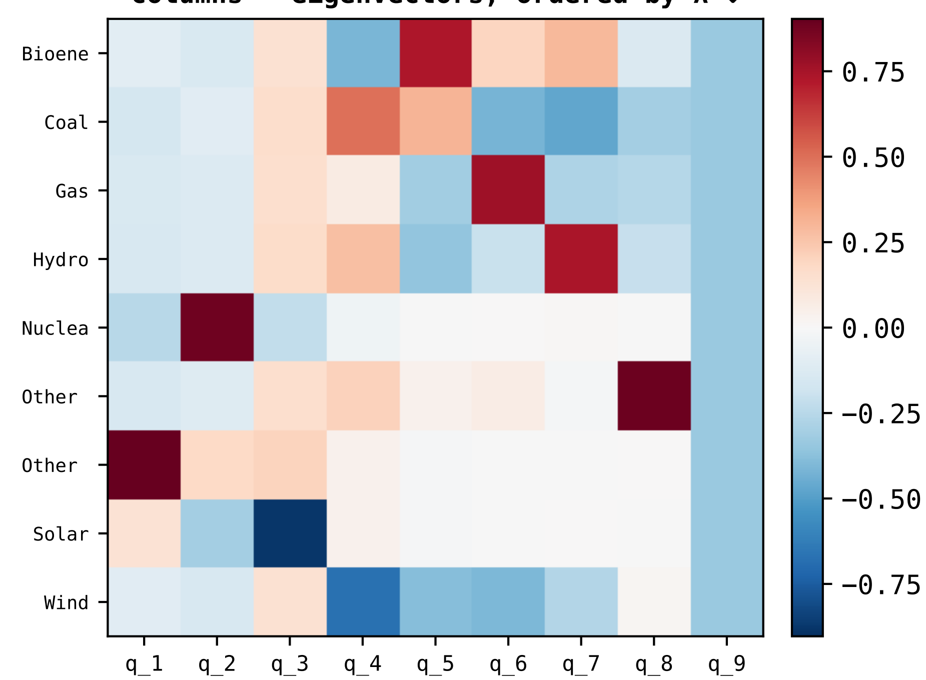
```
raw composition
     │  closure C(x) = x / Σx
     ▼
closed composition x ∈ Δ^{D−1}
     │  CLR: x → log x − ⟨log x⟩
     ▼
CLR coordinates (R^9)
     │  ILR: clr → clr @ H^T
     ▼
ILR coordinates (R^8)

·  D = 9 carriers
·  N = 26 timesteps
·  each arrow is exact
   (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 213.246 · trace = 352.580



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓



§5 Spectral Theorem verification

```
Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 1.14e-13
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 1.11e-15

Rank-k variance explained:
  k=1  →  60.5%
  k=2  →  90.4%
  k=3  →  99.9%
  k=9  →  100.0%
```

Foundations — Numeric Verification — EMB DEU Germany generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	2.222e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	213.2461	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	352.5800	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	1.137e-13	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	1.110e-15	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	60.48%	—
§6 Decomp	cumulative variance at k=2	90.42%	—
§6 Decomp	cumulative variance at k=3	99.92%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

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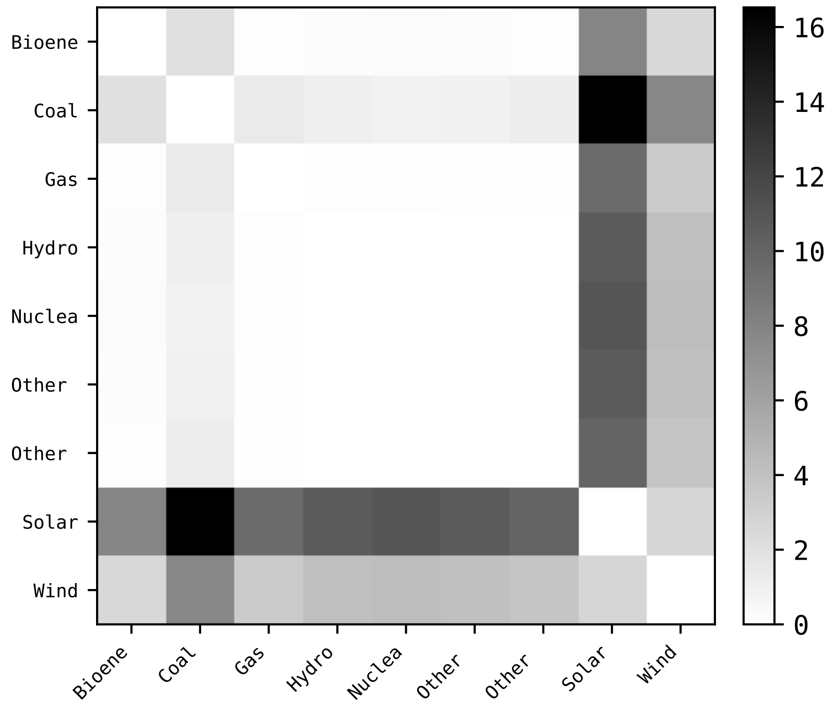
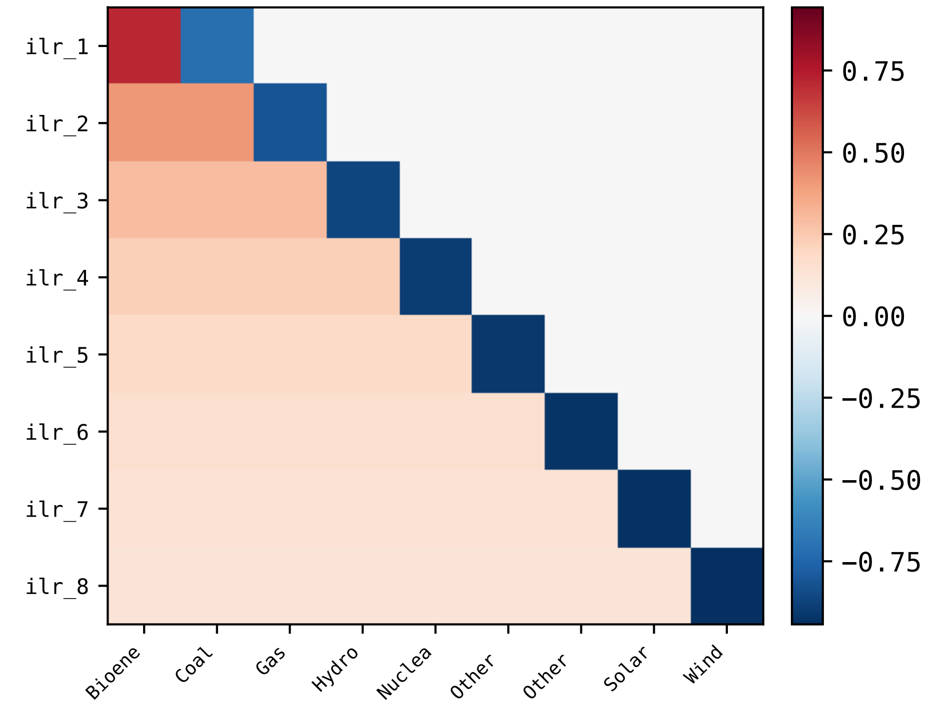
Foundations Plate (Stage 0) — EMB FRA France generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

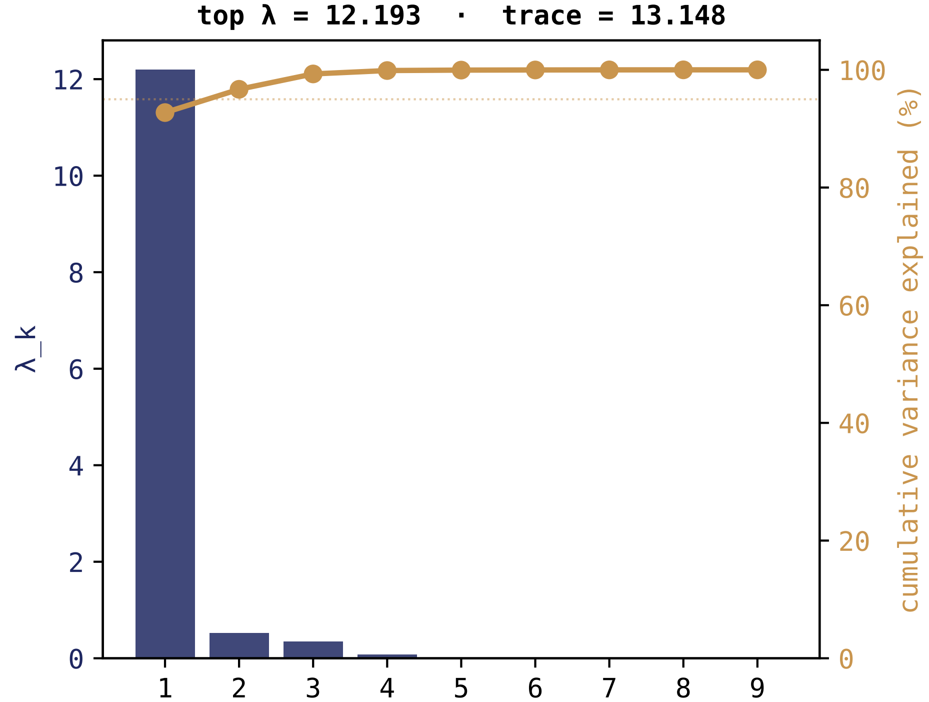


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

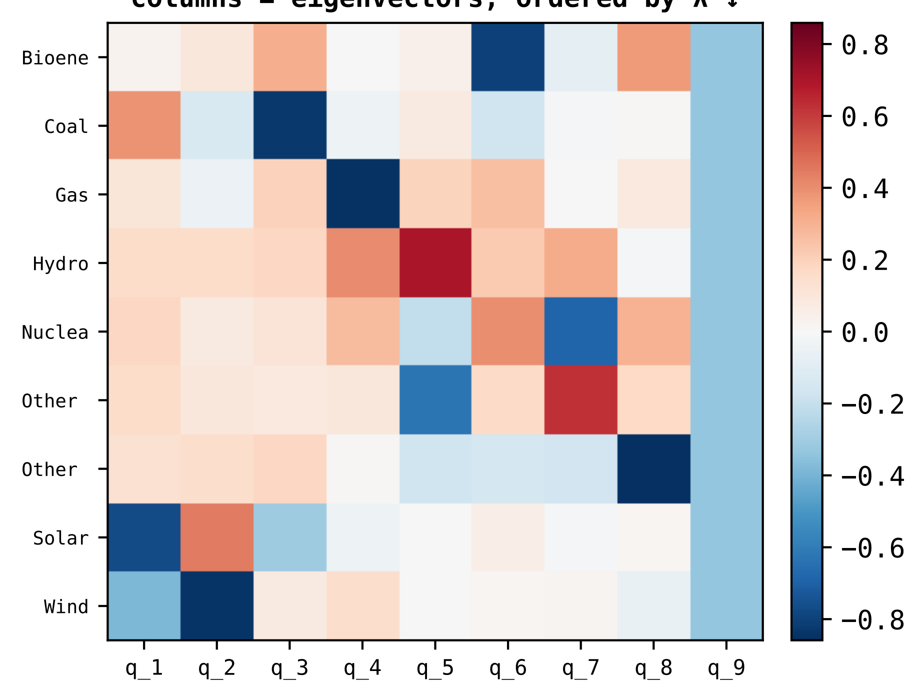



```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^9)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^8)

  ·  D = 9 carriers
  ·  N = 26 timesteps
  ·  each arrow is exact
     (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 12.193 · trace = 13.148



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓




```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 4.44e-15
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 1.33e-15

Rank-k variance explained:
  k=1  →  92.7%
  k=2  →  96.7%
  k=3  →  99.3%
  k=9  →  100.0%
```

Foundations — Numeric Verification — EMB FRA France generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	2.222e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	12.1929	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	13.1476	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	4.441e-15	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	1.332e-15	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	92.74%	—
§6 Decomp	cumulative variance at k=2	96.69%	—
§6 Decomp	cumulative variance at k=3	99.30%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
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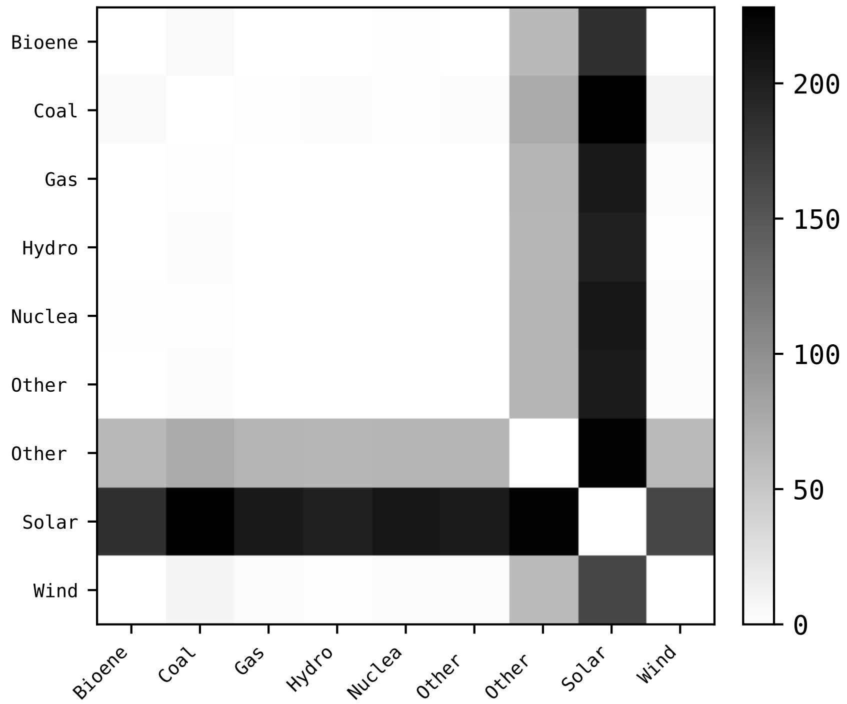
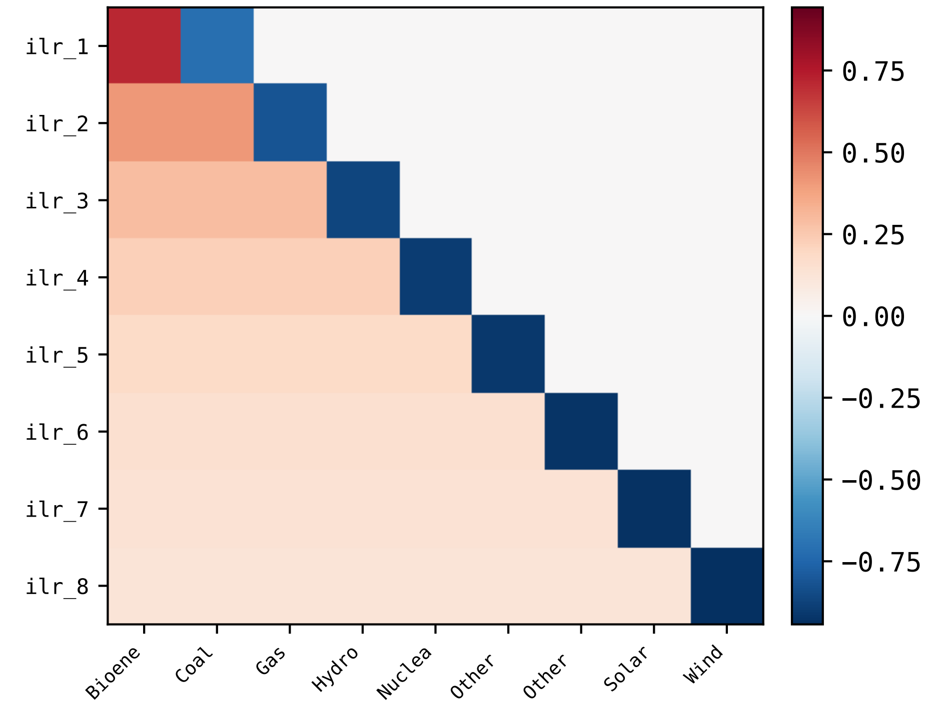
Foundations Plate (Stage 0) — EMB GBR United Kingdom generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

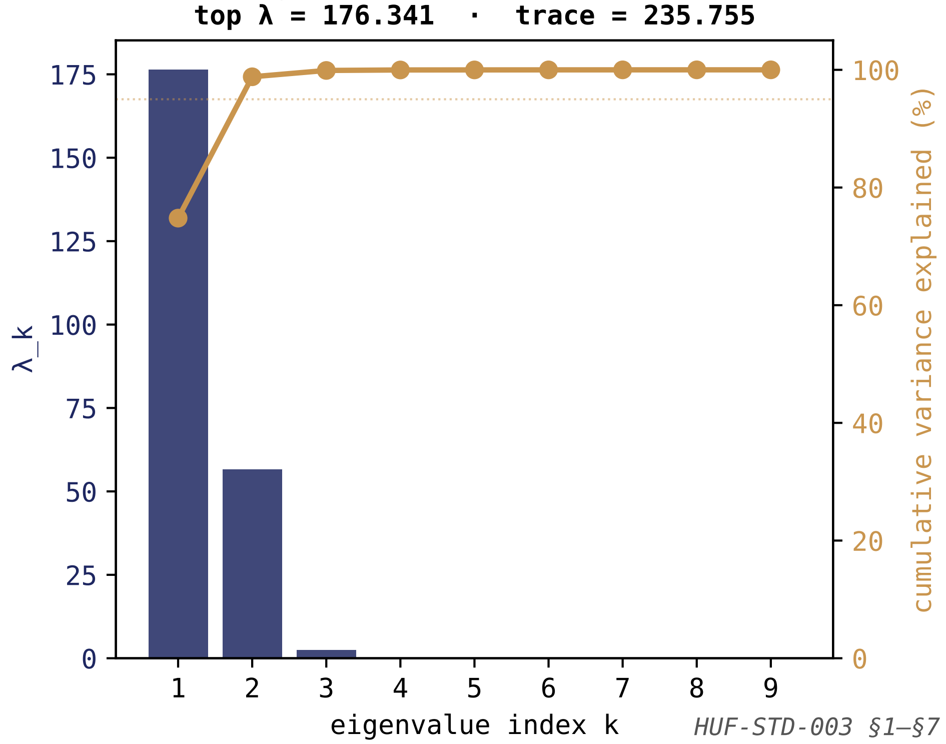


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

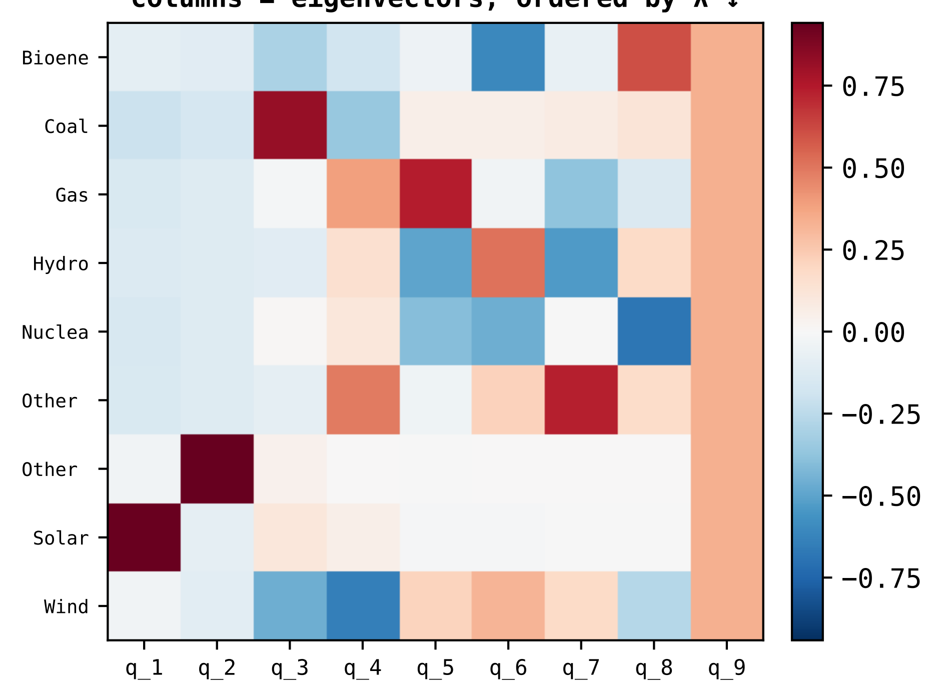



```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^9)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^8)

 ·  D = 9 carriers
 ·  N = 26 timesteps
 ·  each arrow is exact
    (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 176.341 · trace = 235.755



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓




```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 7.11e-14
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 8.88e-16

Rank-k variance explained:
  k=1  →  74.8%
  k=2  →  98.8%
  k=3  →  99.9%
  k=9  →  100.0%
```

Foundations — Numeric Verification — EMB GBR United Kingdom generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	$\max\lvert M - M^T\rvert$ for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	$\max\lvert \Sigma - \Sigma^T\rvert$ for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	$\max\lvert H H^T - I_{D-1}\rvert$ (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	$\max\lvert H^T H - P_{D-1}\rvert$ (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	2.222e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	$\max\lvert \mathrm{imag}(\lambda_k)\rvert$ (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	176.3412	—
§4 Eigen	$\mathrm{trace}(\Sigma) = \Sigma\, \lambda_k$ (total variance)	235.7552	—
§5 Spectral	$\max\lvert \Sigma - Q \Lambda Q^T\rvert$ (Spectral Theorem residual)	7.105e-14	IEEE-floor
§5 Spectral	$\max\lvert Q^T Q - I\rvert$ (orthonormality of eigenbasis)	8.882e-16	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	74.80%	—
§6 Decomp	cumulative variance at k=2	98.82%	—
§6 Decomp	cumulative variance at k=3	99.90%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

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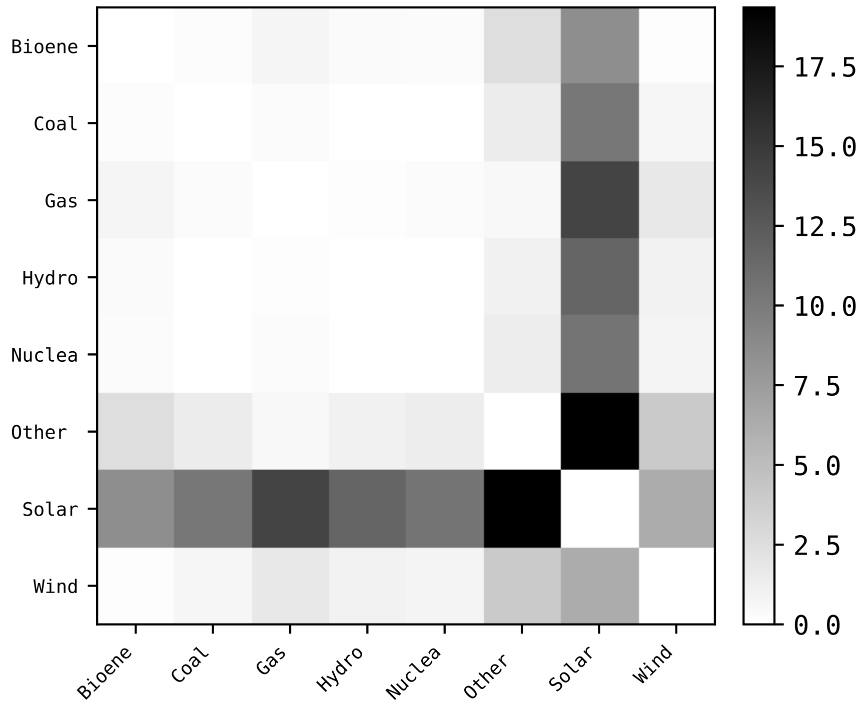
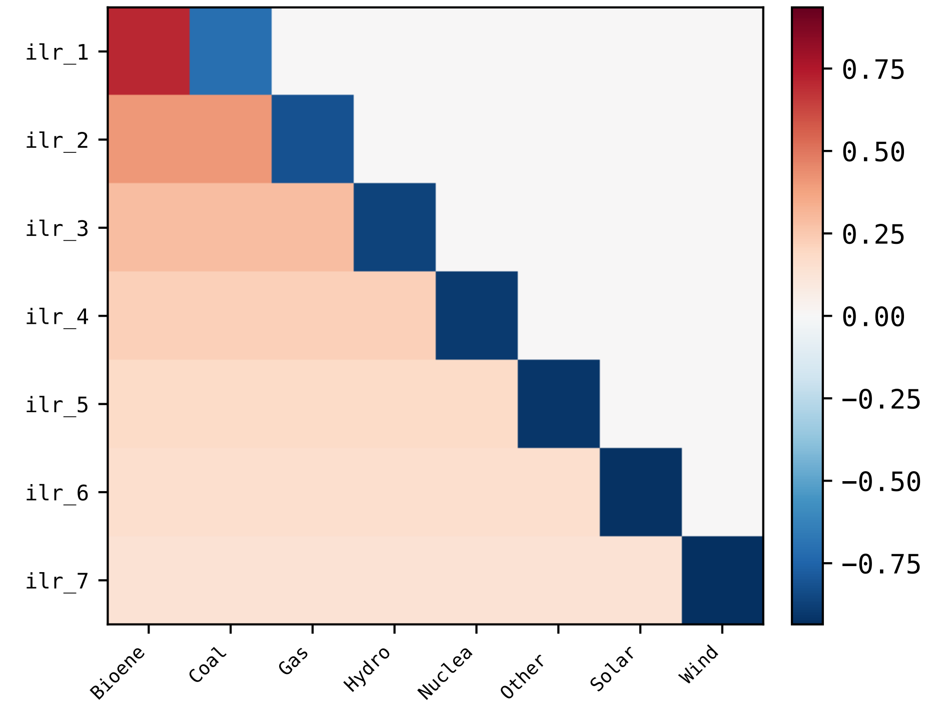
Foundations Plate (Stage 0) — EMB IND India generation TWh.csv · D = 8 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

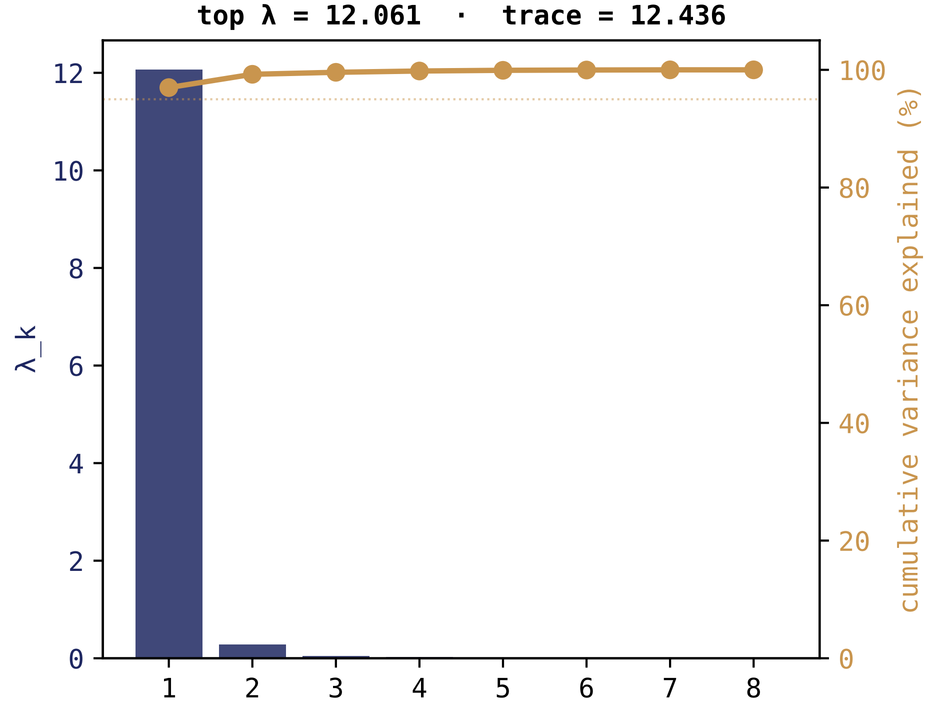


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

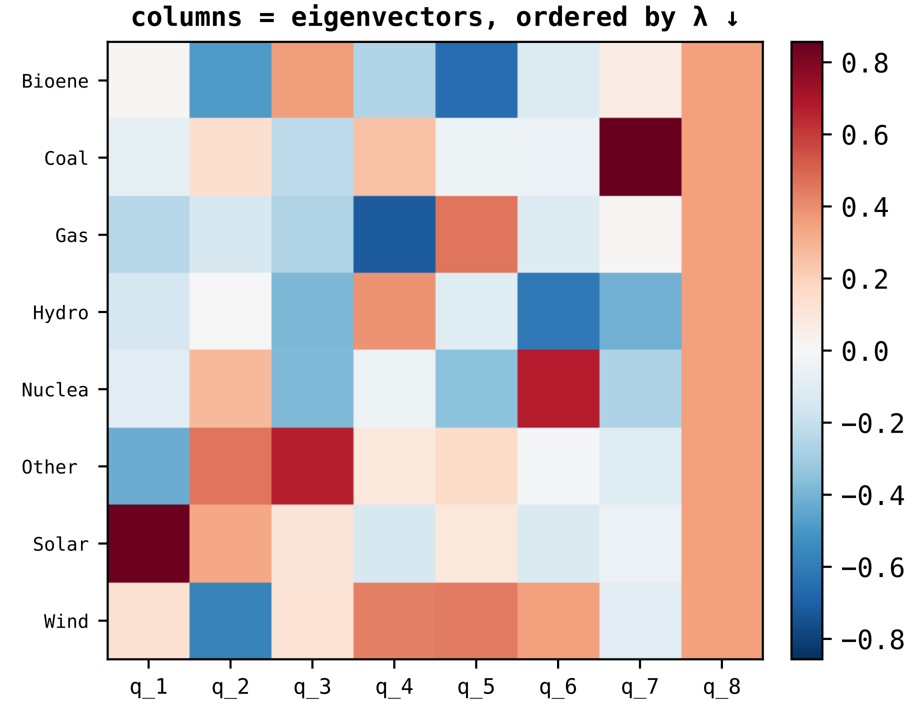



```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^8)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^7)

  ·  D = 8 carriers
  ·  N = 26 timesteps
  ·  each arrow is exact
     (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 12.061 · trace = 12.436



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓




```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 1.07e-14
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 8.88e-16

Rank-k variance explained:
  k=1  →  97.0%
  k=2  →  99.2%
  k=3  →  99.6%
  k=8  →  100.0%
```

Foundations — Numeric Verification — EMB IND India generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

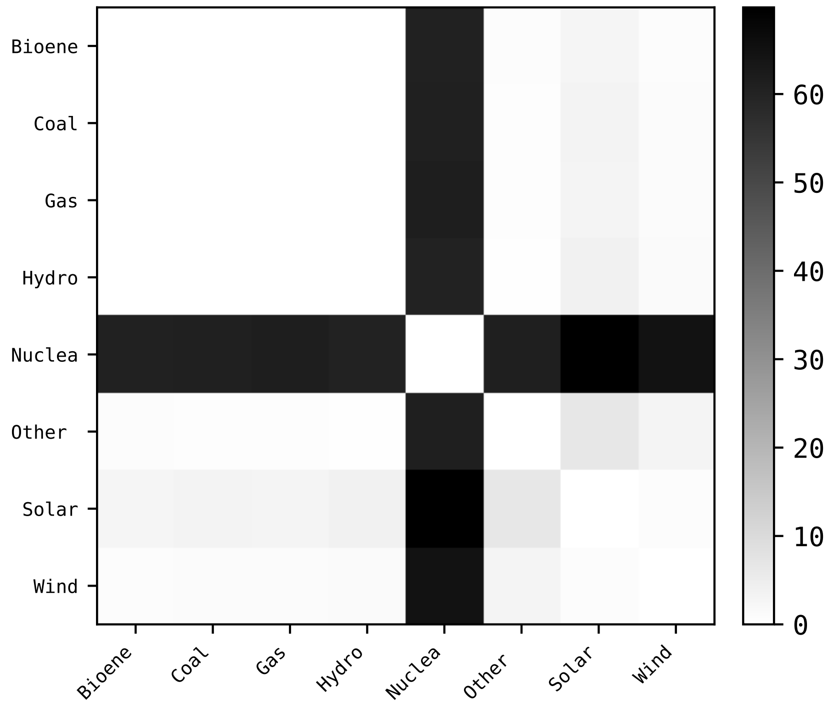
FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	$\max\lvert M - M^T\rvert$ for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	$\max\lvert \Sigma - \Sigma^T\rvert$ for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	$\max\lvert H H^T - I_{D-1}\rvert$ (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	$\max\lvert H^T H - P_{D-1}\rvert$ (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	3.750e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	8	= D (8)
§4 Eigen	$\max\lvert \mathrm{imag}(\lambda_k)\rvert$ (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	12.0611	—
§4 Eigen	$\mathrm{trace}(\Sigma) = \Sigma\ \lambda_k$ (total variance)	12.4363	—
§5 Spectral	$\max\lvert \Sigma - Q \Lambda Q^T\rvert$ (Spectral Theorem residual)	1.066e-14	IEEE-floor
§5 Spectral	$\max\lvert Q^T Q - I\rvert$ (orthonormality of eigenbasis)	8.882e-16	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	96.98%	—
§6 Decomp	cumulative variance at k=2	99.24%	—
§6 Decomp	cumulative variance at k=3	99.58%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.

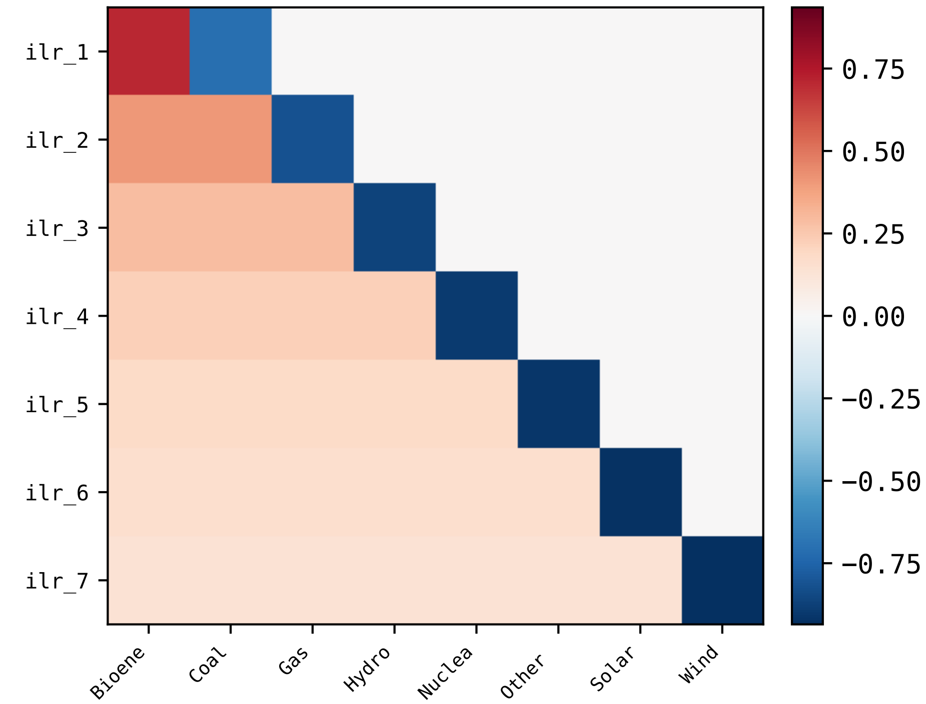
Foundations Plate (Stage 0) — EMB JPN Japan generation TWh.csv · D = 8 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

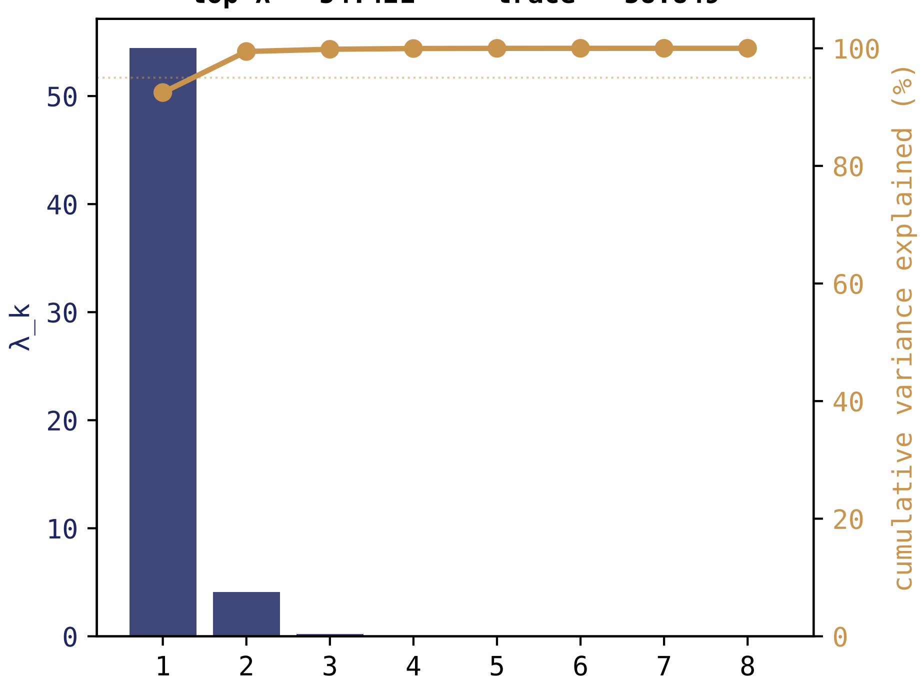


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16



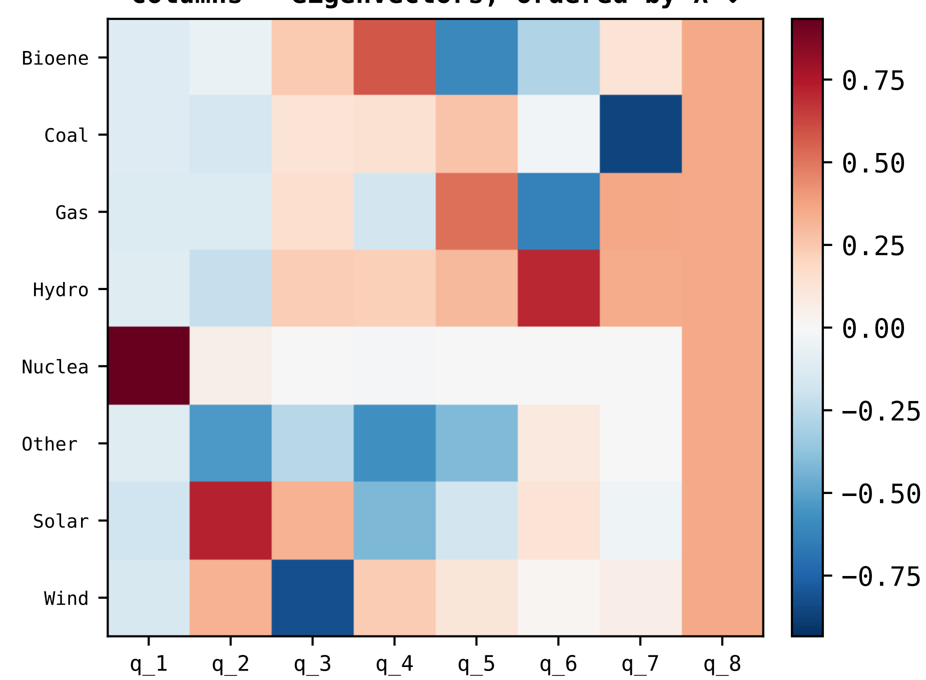

```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^8)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^7)

  ·  D = 8 carriers
  ·  N = 26 timesteps
  ·  each arrow is exact
     (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 54.421 · trace = 58.849



eigenvalue index k

§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓




```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 6.39e-14
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 8.88e-16

Rank-k variance explained:
  k=1  →  92.5%
  k=2  →  99.5%
  k=3  →  99.8%
  k=8  →  100.0%
```

Foundations — Numeric Verification — EMB JPN Japan generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

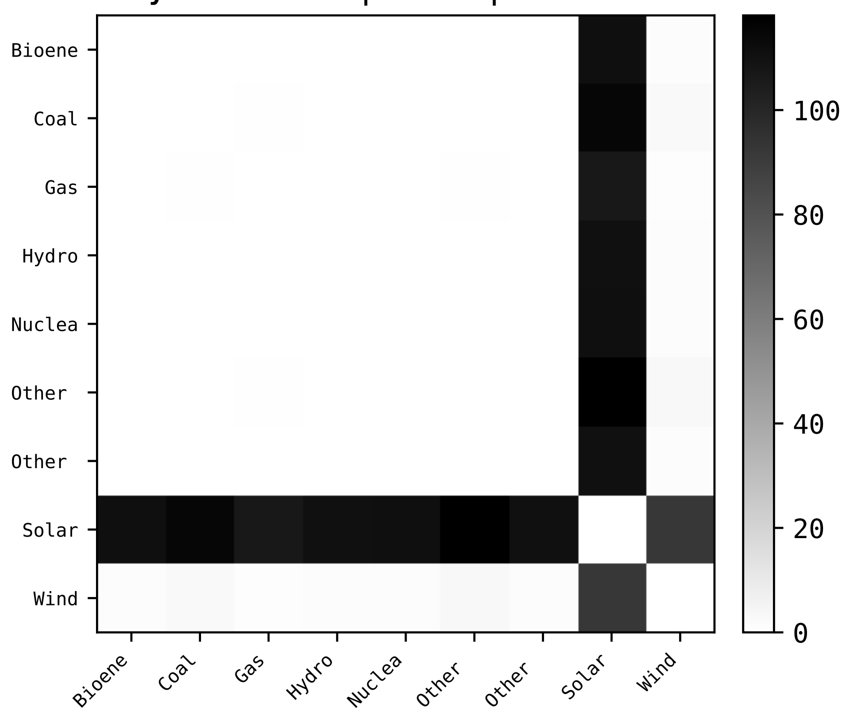
FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	2.500e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	8	= D (8)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	54.4206	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	58.8490	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	6.395e-14	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	8.882e-16	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	92.47%	—
§6 Decomp	cumulative variance at k=2	99.46%	—
§6 Decomp	cumulative variance at k=3	99.84%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.

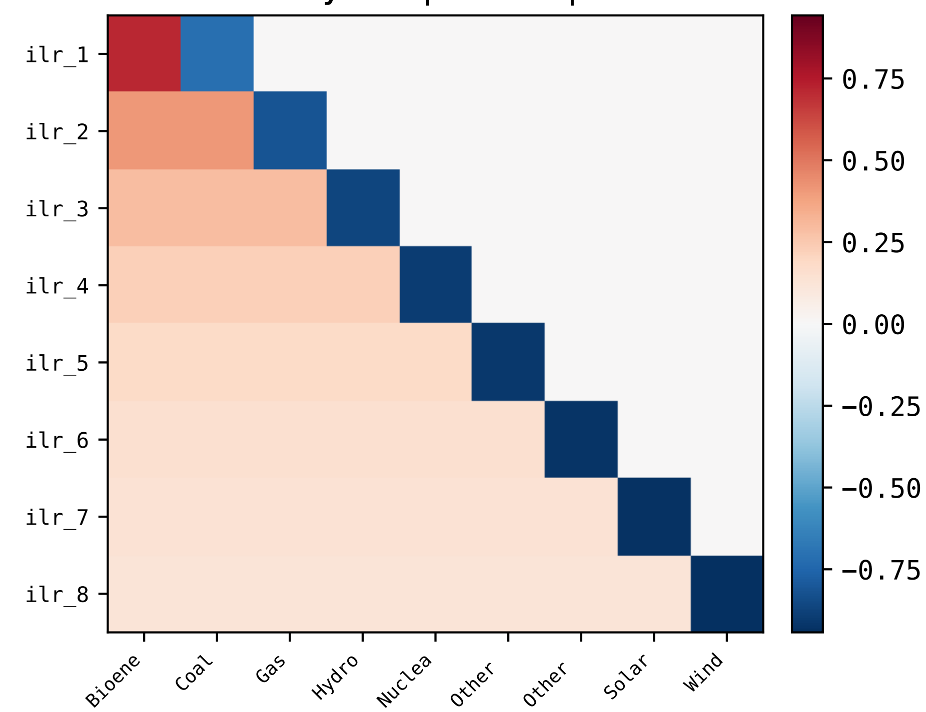
Foundations Plate (Stage 0) — EMB USA United States generation TWh.csv · D = 9 carriers · N = 25 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

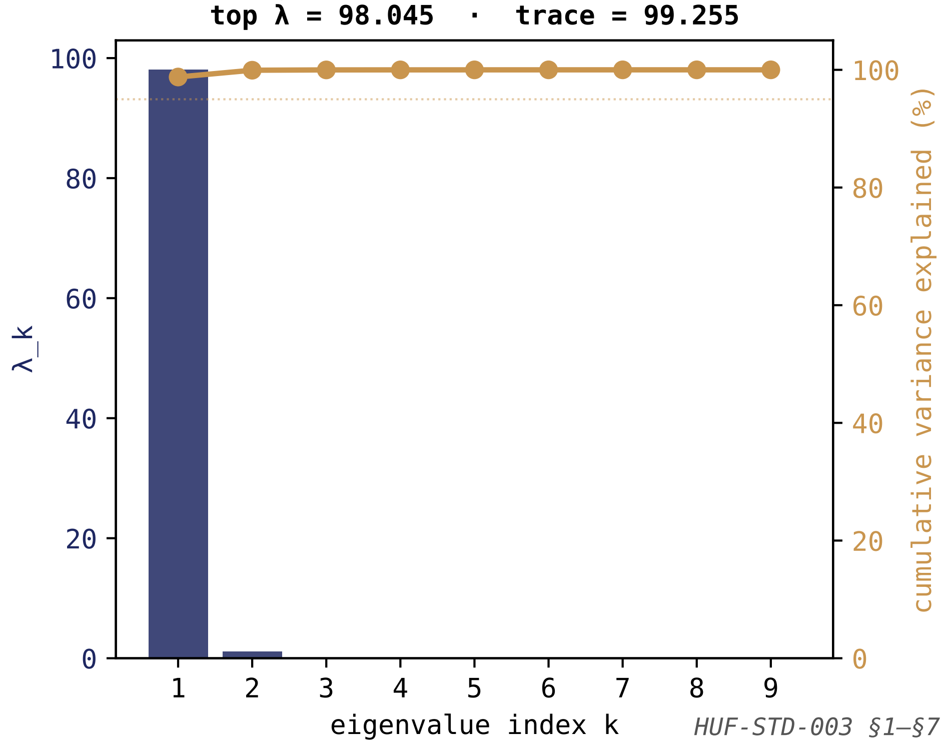


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16




```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^9)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^8)

  ·  D = 9 carriers
  ·  N = 25 timesteps
  ·  each arrow is exact
     (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 98.045 · trace = 99.255



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓


```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 5.68e-14
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 1.78e-15

Rank-k variance explained:
  k=1  →  98.8%
  k=2  →  99.9%
  k=3  →  100.0%
  k=9  →  100.0%
```

Foundations — Numeric Verification — EMB USA United States generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

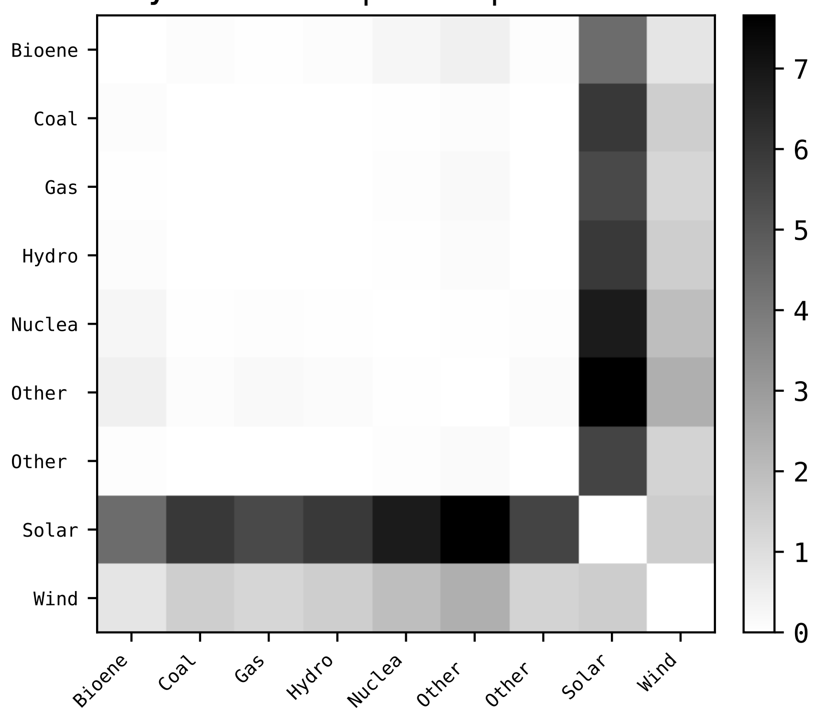
FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	max|M − M^T| for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	max|Σ − Σ^T| for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	max|H H^T − I_{D−1}| (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	max|H^T H − P_{D-1}| (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	2.222e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	max|imag(λ_k)| (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	98.0448	—
§4 Eigen	trace(Σ) = Σ λ_k (total variance)	99.2549	—
§5 Spectral	max|Σ − Q Λ Q^T| (Spectral Theorem residual)	5.684e-14	IEEE-floor
§5 Spectral	max|Q^T Q − I| (orthonormality of eigenbasis)	1.776e-15	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	98.78%	—
§6 Decomp	cumulative variance at k=2	99.93%	—
§6 Decomp	cumulative variance at k=3	99.99%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

The foundations carry the bedrock. The instrument reads. The expert decides.
The hashes carry the receipts. The vocabulary holds the line.

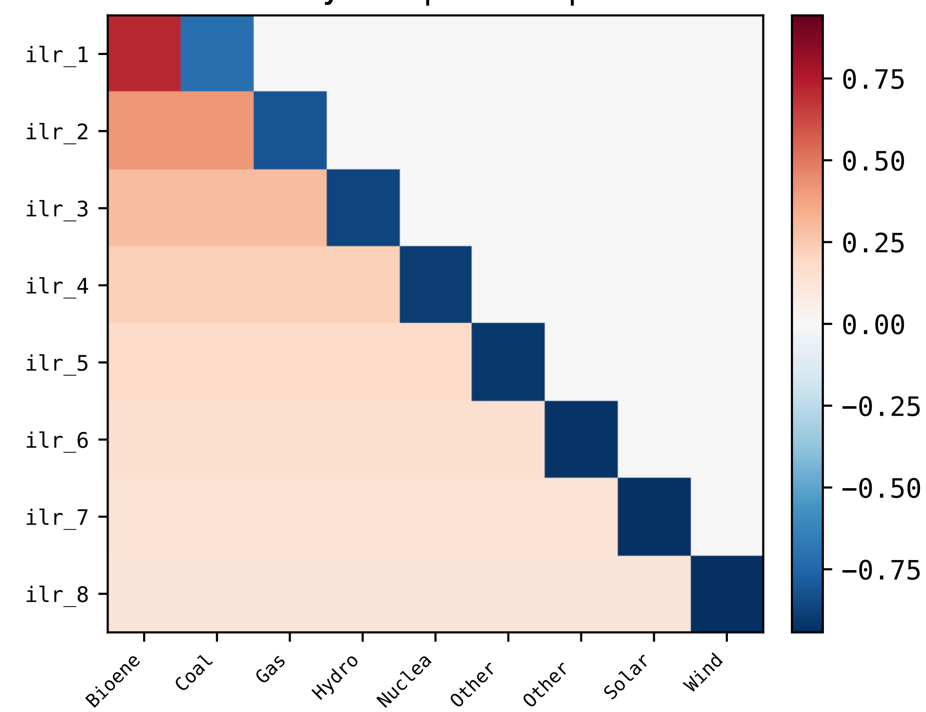
Foundations Plate (Stage 0) — EMB WLD World generation TWh.csv · D = 9 carriers · N = 26 timesteps

Seven linear-algebra foundations of Hs per HUF-STD-003 · the bedrock made visible

§1 Variation matrix var(log x_i/x_j)
symmetric: max|M − M^T| = 0.00e+00

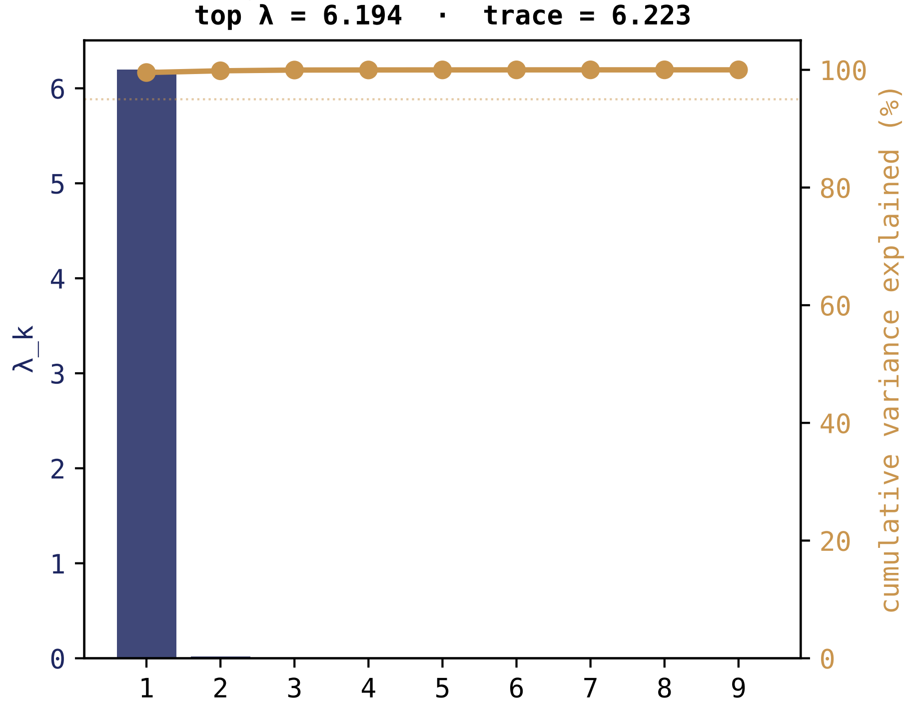


§2 Helmert basis H (D−1 × D)
orthonormality: max|HH^T − I| = 2.22e-16

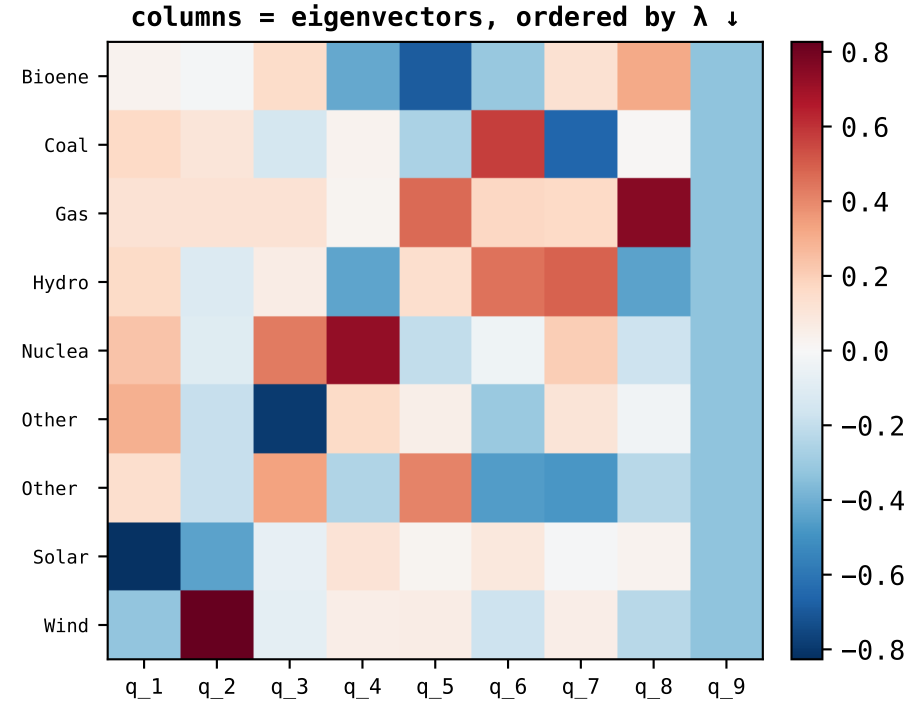



```
§3  Matrix Decomposition Chain

   raw composition
        │  closure C(x) = x / Σx
        ▼
   closed composition x ∈ Δ^{D−1}
        │  CLR: x → log x − ⟨log x⟩
        ▼
   CLR coordinates (R^9)
        │  ILR: clr → clr @ H^T
        ▼
   ILR coordinates (R^8)

  ·  D = 9 carriers
  ·  N = 26 timesteps
  ·  each arrow is exact
     (modulo IEEE float)
```

§4 Eigenvalue scree (CLR covariance)
top λ = 6.194 · trace = 6.223



§6 Orthonormal eigenbasis Q
columns = eigenvectors, ordered by λ ↓




```
§5  Spectral Theorem verification

Σ = Q Λ Q^T  (theorem)
max|Σ − Q Λ Q^T| = 8.88e-16
  (IEEE-floor → theorem holds)

Σ symmetric?
max|Σ − Σ^T| = 0.00e+00

All λ real?
max|imag(λ)| = 0.00e+00  (eigh ⇒ real)

Q orthonormal?
max|Q^T Q − I| = 1.33e-15

Rank-k variance explained:
  k=1  →  99.5%
  k=2  →  99.8%
  k=3  →  100.0%
  k=9  →  100.0%
```

Foundations — Numeric Verification — EMB WLD World generation TWh.csv

Machine-precision proof that the seven foundations hold on the actual data

FOUNDATION	STATEMENT	VALUE	JUDGMENT
§1 Symmetric	$\max\lvert M - M^T\rvert$ for variation matrix	0.000e+00	exact (= 0)
§1 Symmetric	$\max\lvert \Sigma - \Sigma^T\rvert$ for CLR covariance	0.000e+00	IEEE-floor (2.22e-16)
§2 Transpose	$\max\lvert H H^T - I_{D-1}\rvert$ (Helmert orthonormal rows)	2.220e-16	IEEE-floor
§2 Transpose	$\max\lvert H^T H - P_{D-1}\rvert$ (projection onto image)	2.220e-16	IEEE-floor
§3 Decomp	closure → CLR → ILR → CLR roundtrip residual	1.111e-09	IEEE-floor
§4 Eigen	number of eigenvalues found	9	= D (9)
§4 Eigen	$\max\lvert \mathrm{imag}(\lambda_k)\rvert$ (real because Σ symmetric)	0.000e+00	exact
§4 Eigen	λ_1 (top eigenvalue, variance along leading axis)	6.1944	—
§4 Eigen	$\mathrm{trace}(\Sigma) = \Sigma\, \lambda_k$ (total variance)	6.2228	—
§5 Spectral	$\max\lvert \Sigma - Q \Lambda Q^T\rvert$ (Spectral Theorem residual)	8.882e-16	IEEE-floor
§5 Spectral	$\max\lvert Q^T Q - I\rvert$ (orthonormality of eigenbasis)	1.332e-15	IEEE-floor
§6 Decomp	cumulative variance at k=1 (rank-1 approximation)	99.54%	—
§6 Decomp	cumulative variance at k=2	99.84%	—
§6 Decomp	cumulative variance at k=3	99.97%	—
§7 Visualize	Stage-0 Foundations Plate (page 1 of this PDF)	present	delivered

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